△ Flow : • Flow rate

△ Basic volume : • Tidal volume (TV)
• Inspiratory reserve volume (IRV)
• Expiratory reserve volume (ERV)
• Residual volume (RV)

△ Functional capacity : • Inspiratory capacity (IC)
• Functional residual capacity (FRC)
• Vital capacity (VC)
• Total lung capacity (TLC)

• Introduction - Circulation

△ Cardiac output (CO) :

$CO = SV \times HR$, where SV denotes stroke volume, and HR heart rate.

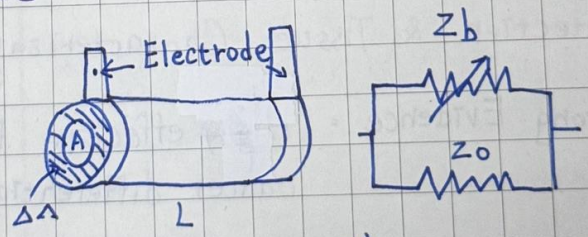
△ Methods :

Oxygen Fick method, Indicator dilution method, etc.

• Bioimpedance Method

Basic model :

$$R = \frac{V}{I} = \rho \frac{L}{A}$$



$$Z_b = \rho_b \frac{L}{\Delta A} \Rightarrow \Delta Vol = L \Delta A = \rho_b \frac{L^2}{Z_b}$$

$$\Delta Z = Z_0 - \frac{Z_0 Z_b}{Z_0 + Z_b} \Rightarrow \Delta Vol = \rho_b \frac{L^2 \Delta Z}{Z_0^2}$$

$$SV = \left(\rho_b \frac{L^2}{Z_0^2}\right) \cdot LVET \cdot \left(\frac{dZ}{dt}\right)_{max}$$

LVET = Left Ventricle Ejection Time

- Fundamental Application

Bioimpedance is a versatile, non-invasive tool used to measure the electical propeties of biological tissues. It is primarily used to track:

- Volume changes: Including Impedance Plethysmography (IPG) and Impedance Cardiography (ICG).
- Body Composition: Analyzing fat, musle, and water distribution
- Tissue Characterization: Differentiating between normal, pre-malignant, and malignant tissues (e.g., cancer detection in cervical, prostate, and breast tissues).

- Impedance Cardiography (ICG)

- Mechanism: A high-frequency (100kHz) constant current is applied to the thorax.
- Data Output: It tracks changes in thoracic impedance (Z_0, ΔZ, and the derivative dZ/dt) alongside ECG and phonocardiograms to calculate how mach blood the heart pumps.

- Cancer Detection & Tissue Characterization

- Strong Evidence: Highy effective for Cervical and Prostate cancer differentiation.
- Moderate Evidence: Useful for Breast, Skin, and Bladder cancers.
- Method: It identifies the unique "electrical signature" of malignant cells compared to normal tissue.

- Electrical Impedance Tomography (EIT)

EIT is an emerging imaging modality that maps internal conductivity.

  - Key Advantage : Unlike X-rays or CT scans, EIT is radiation-free and provides real-time imaging.
  - Primary Use : Monitoring lung ventilation and circulatory functions at the bedside.

- Electrical Impedance Myography (EIM)

  - Clinical Value : It helps physicians evaluate disease severity, track progression over time, and measure how a patient responds to therapy.
  - Visualizing Data : Uses Cole-plots to determine muscle health.

- Comparison of Imaging Modalities

Feature	EIT	CT	Ultrasound	MRI
Principle	Impedance	X-rays	Sound Waves	Radio Waves
Cost	Low	Moderate	Low	Very High
Portability	Portable	Non-portable	Portable	Non-portable
Radiation	None	Ionizing	None	None
Status	In Development	Mature	Mature	Mature

- EIT Algorithms & Clinical Practice

1. Image Reconstruction (The Math)

The core challenge of EIT is solving the Poisson equation with boundary conditions.

$$\nabla \cdot \sigma \nabla V = 0 \text{ in } \Omega$$

- Forward Problem: Given internal conductivity (σ), find the boundary voltages (V).
- Inverse Problem: Given measured voltages (V), reconstrust the internal conductivity (σ) to create the image.
- Solvers: Methods range from traditional math (Backprojection, FEM) to modern Neural Networks (Deep learning) for faster, clearer results.

2. Clinical Systems

EIT is used at the bedside for real-time monitoring:

- Key Systems: PulmoVista 500 (Dräger) and Enlight 1800 (Timpel).
- Integration: Often intergrated directly with ventilators (e.g. Elisa 800 VIT) to help doctors see the immediate effect of breathing treatments.

- Medical Imaging Fundamentals
  - CT Baseline: Works by measuring beam attenuation from many angles to reconstruct images.
  - System Essentials: All imaging requires an energy source, tissue interaction, detectability, and a reconstruction algorithm.

- EIT System Design
  - Hardware: Uses an electrode array (2 or 4-electrode setups) and a control circuit to apply current/voltage.
  - Workflow: Signal → Measurement → Computer (Reconstruction Algorithm) → Display.
  - Processing: Uses high-speed components (like the ADSP 2101) for real-time data acquisition.

- Commercial Status
  - Available Systems: Palmovista (Dräger), Goe-MF II, and Enlight.
  - Main Application: Currently used for real-time bedside lung ventilation monitoring.

3. Lung Function Applications

EIT is primarily used to characterize lung behavior:

- Regional Ventilation: Seeing exactly which parts of the lungs are receiving air.
- Optimal Settings: Helping doctors set the correct PEEP (Positive End-Expiratory Pressure) to prevent lung collapse or over-inflation
- Dynamic Monitoring: Real-time tracking of lung transplants, pleural effusions, and physiotherapy progress.